CS & IT ENGINEERING

DISCRETE MATHS
SET THEORY

Lecture No. 10

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TOPICS TO BE COVERED

01 Basics of relations

02 Types of relations

03 Number Of relations

Relations

$A \to$ non empty set.

$R_1 \to$ equivalence Relation $R_2 \to$ equivalence Relation.

$\underline{R_1 \cup R_2 =}$

$$\begin{array}{cccc} R_1 & R_1 & S_1 & T_1 \\ & \cup & \cup & \cup \\ R_2 & R_2 & S_2 & T_2 \\ & R & S & \end{array}$$

$\underline{R_1 \cap R_2}$ = equivalence Relation

$$\begin{array}{cccc} R_1: & R & S & T \\ & \cap & \cap & \cap \\ R_2: & R & S & T \end{array}$$

Relations

	Union	Intersection
R_1, R_2 equivalence Relation	✗	✓

Relations

$R_1 = \{(1\,1)(2\,2)(3\,3)(1\,2)(2\,1)\}$ R S T

$R_1 \cap R_2 = \{(1\,1)(2\,2)(3\,3)\}$

$R_2 = \{(1\,1)(2\,2)(3\,3)(2\,3)(3\,2)\}$ R S T

$R_1 \cup R_2 = \{(1\,1)(2\,2)(3\,3)(1\,2)(2\,1)(2\,3)(3\,2)\}$

13 is absent

Partial Order Relation

@ all same time

- Reflexive ✓
- Antisymmetric ✓
- Transitive

(RAT)

Relations

Domain: Z^+

$R_1 = \{(a, b) \mid a \text{ divides } b\}$

$R_1 = \{(a, b) \mid a|b\}$

a divides b

↓ OR

$a|b$ OR $\frac{b}{a}$

R: aRa $\quad a|a$ ✓

Anti: → no flipping

→ allows same

$aRb \wedge bRa \longrightarrow a=b$

Relations

Anti: $aRb \wedge bRa \rightarrow a=b.$

$a|b \wedge b|a \rightarrow a=b$

$3|6 \wedge 6|3 \longrightarrow 3=6.$

$T \wedge F$

$F \longrightarrow$

True.

$R_1 = \{(3,6)\}$

no flipping

Relations

Transitive: ✓

$aRb \wedge bRc \rightarrow aRc.$

$a|b \wedge b|c \rightarrow a|c.$

$R_2 = \{(a, b) \mid a \leq b\}$ $D: Z$

aRb

R: aRa $a \leq a$ (✓)

A: $aRb \wedge bRa \rightarrow a = b.$

$a \leq b \wedge b \leq a \rightarrow a = b.$

$(1,3) \in R \wedge (3,1) \in R \rightarrow$

$1 \leq 3$

T: $aRb \wedge bRc \rightarrow aRc.$

$a \leq b \wedge b \leq c \rightarrow a \leq c.$

Relations

$(Z^+, |)$ poset

Set : $Z^+ : \{1, 2, 3, 4, 5, \ldots\ldots\}$ (a = 3, b = 5)

3, 7 are not comparable.

Relation : $|$

aRb OR bRa

$3R5$ OR $5R3$

$3|5$ OR $5|3$

3, 5 are not comparable.

eg2: $a = 2 \quad b = 8$

aRb OR bRa.

$2|8$ OR $8|2$

T.

2, 8 are comparable.

Relations

Set : { a, b, c }

(POR) ← Relation : R

if in set only partial elements are comparable then the Relation is called POR

(Partial order Relation)

aRb OR bRa

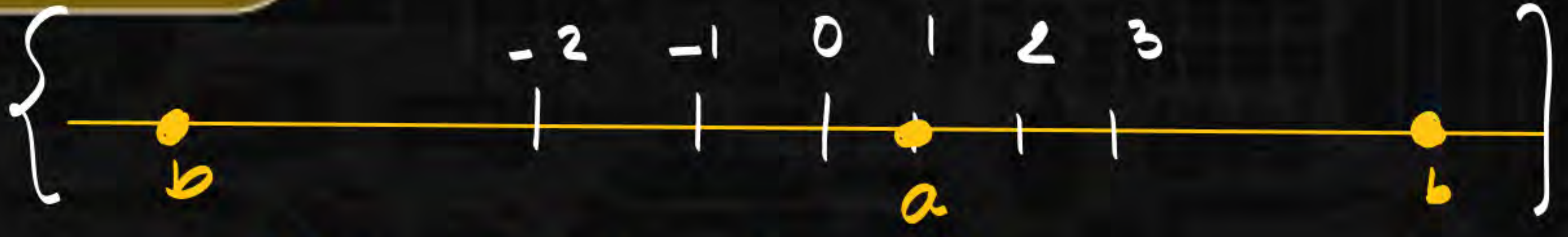
a, b are called comparable.

Otherwise

a, b are not comparable.

Relations

Set: $\mathbb{Z}$.



Relation: $\leq$

eg1: $a = 1$ $b = 10^3$

a, b

$\boxed{aRb \text{ OR } bRa.}$

$a \leq b$ OR $b \leq a$.

$a \leq b$ OR $b \leq a$.

$\underline{1 \leq 10^3}$ OR $\underline{10^3 \leq 1}$.

all elements are comparable in a set

<u>Relation</u>: Total order Relation.
Linear order Relation

Relations

Relation $\longrightarrow$ $\boxed{\text{RAT}} \longrightarrow$ POR $\longrightarrow$ $\boxed{\text{all elements are comparable.}}$

TOR

OR

LOR.

Relations

	union	$\cap$
R_1, R_2 are partial order Relation	X	✓

Relations

1. $R_1 = \{ (1,1)\ (1,2)\ (1,3)\ (1,6)\ (1,12)$

$(2,2)\ (2,6)\ (2,12)$

$(4,4)\ (4,12)$

$(3,3)\ (3,6)\ (3,12)$

$(6,6)\ (6,12)$

$(12,12) \}$ → no flipping → Anti ✓

POR

R

Set: $\{1, 2, 4, 3, 6, 12\}$ → partial order set

(poset)

Relations

$(\underline{\text{Set}}, \underline{\text{Relation}}) \rightarrow$ poset

2. $(\underline{\{1, 2, 3, 4, 6, 12\}}, |) \rightarrow$ poset

POR

3. $(D_{12}, |)$ – poset

D_n : Divisors of n.

D_{12} = Divisors of 12

$= \{1, 2, 3, 4, 6, 12\}$

Relations

$$\left(\underbrace{\text{Set}}_{\text{Toset}},\ \underbrace{\text{Relation}}_{\text{Total ORDER Relation}}\right) \text{ Toset}$$

$(Z^+, |)$ poset

$(Z, \leq)$ Toset

Relations

$R = \{ (1,1)\ (1,2)\ (1,3)\ (1,4)\ (1,6)\ (1,12)$

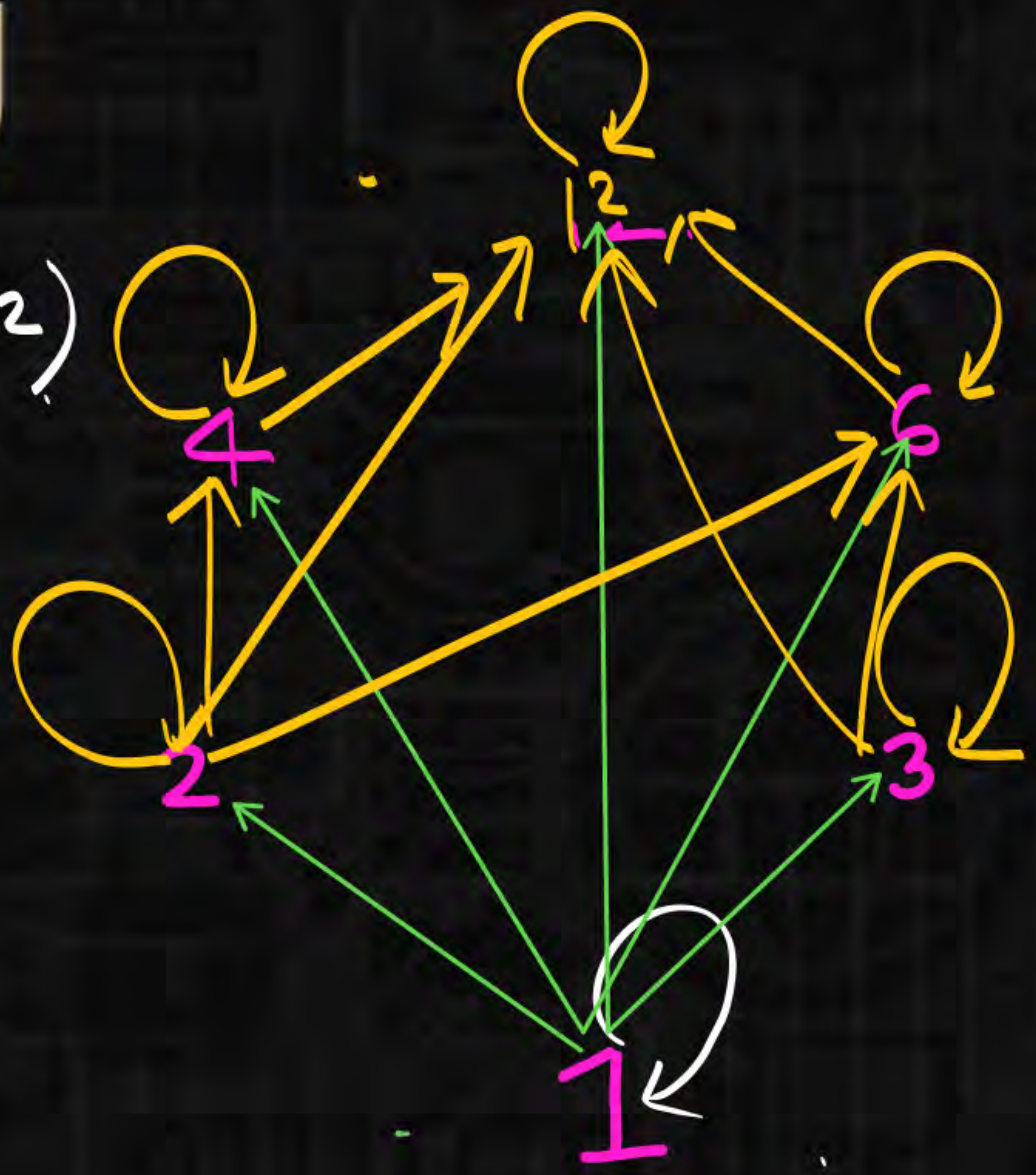
$(2,2)\ (2,4)\ (2,6)\ (2,12)$ ✓

$(3,3)\ (3,6)\ (3,12)$ ✓

$(4,4)\ (4,12)$

$(6,6)\ (6,12)$

$(12,12) \}$



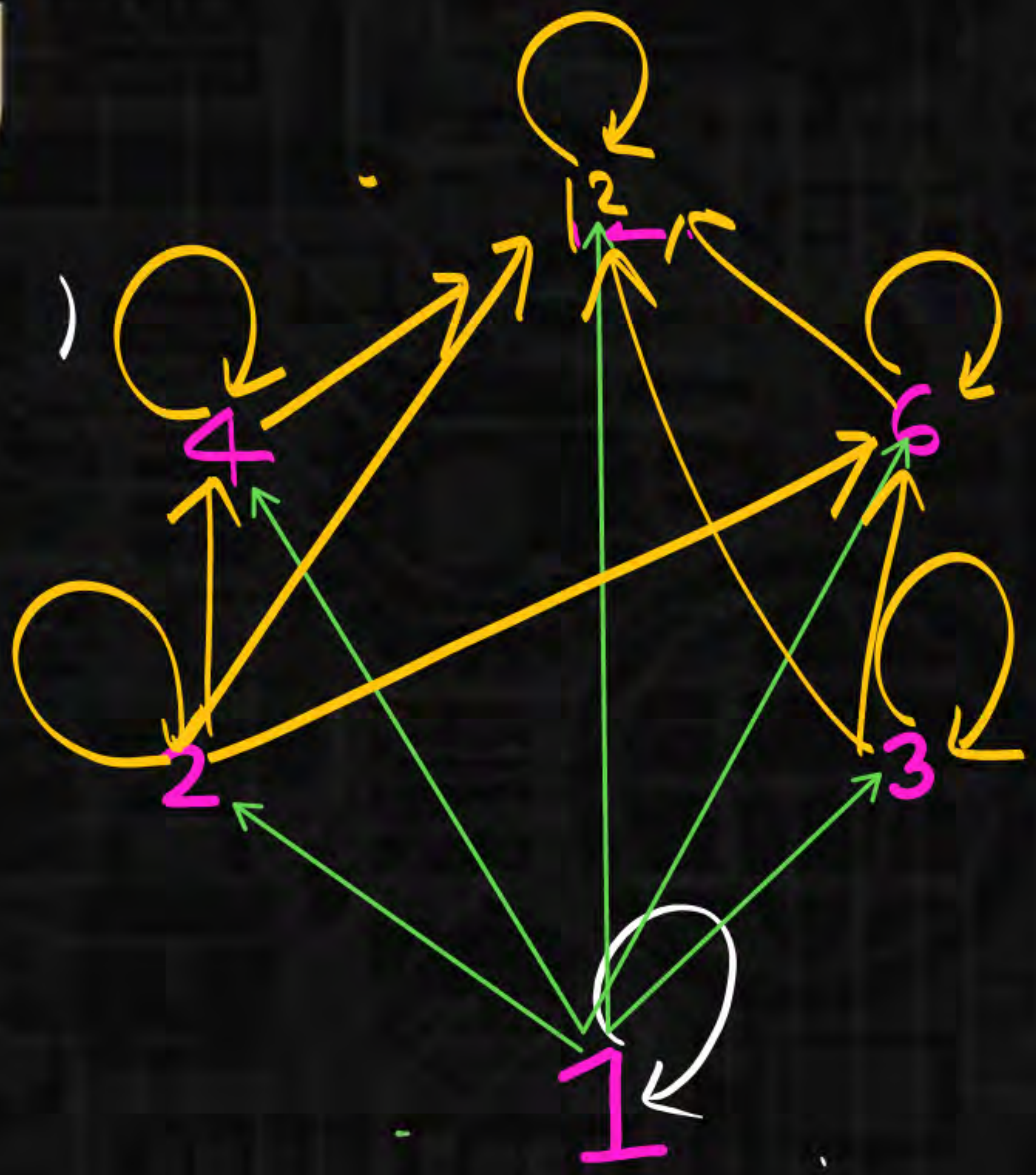
Relations

RAT

Ref : selfloop

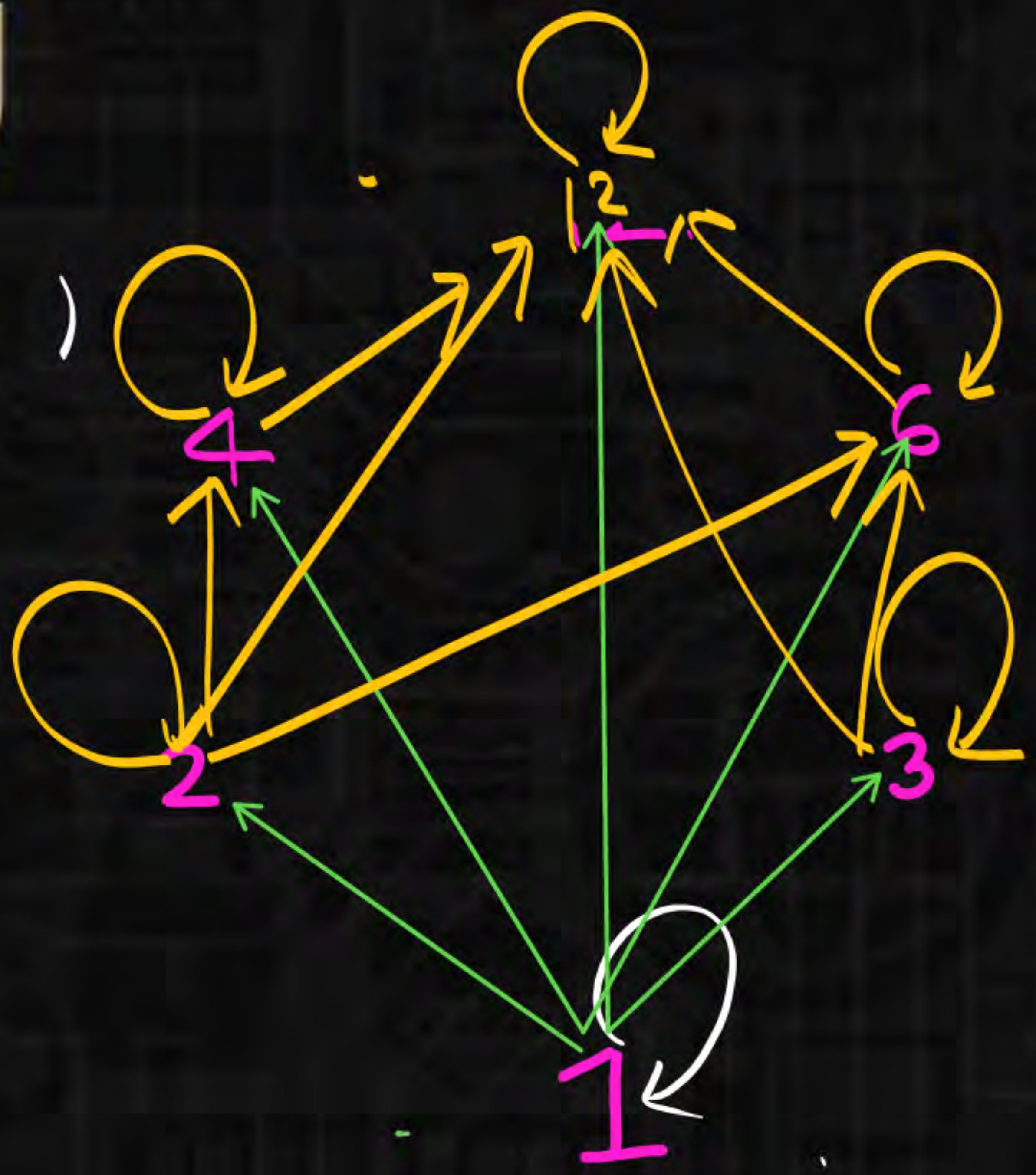
Anti : one direction

Transiti :

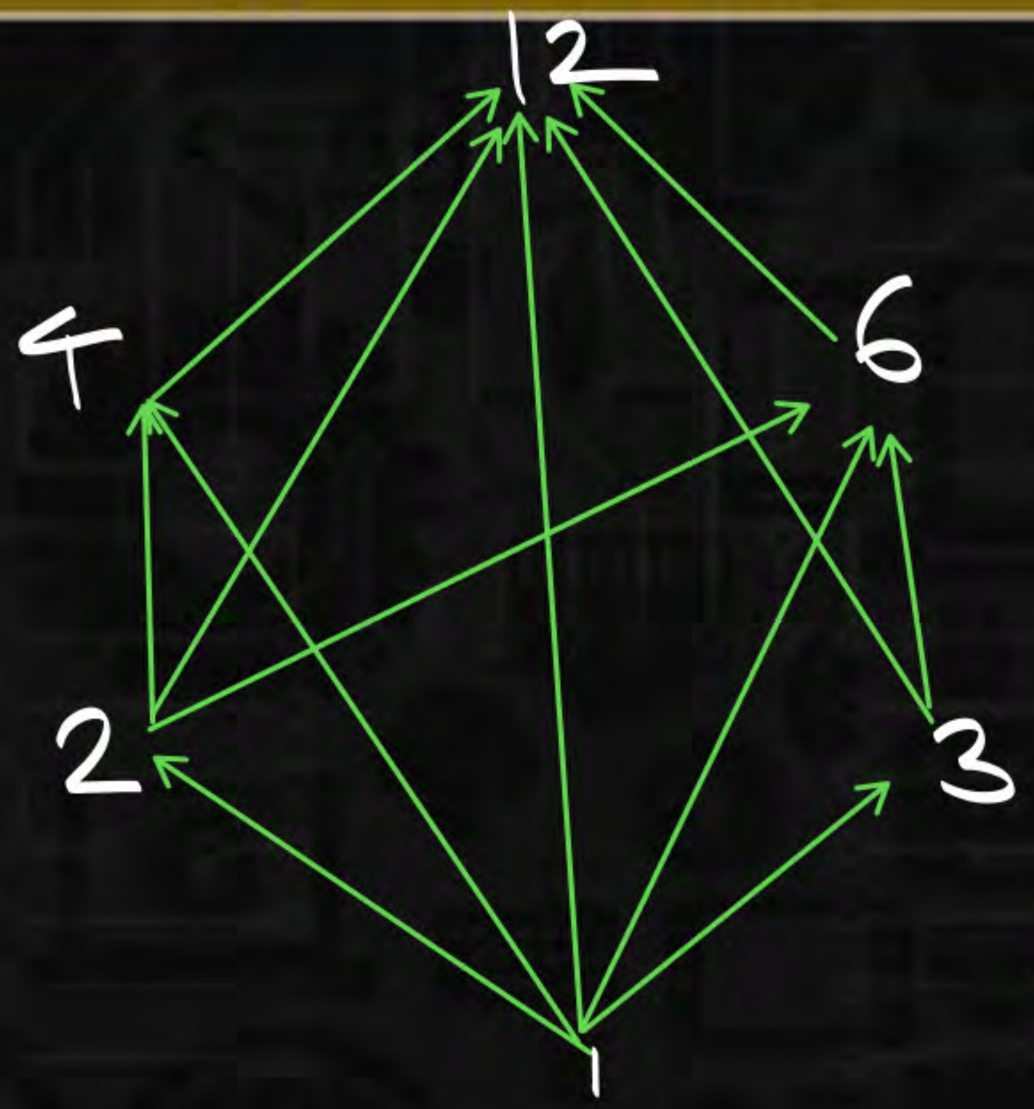


Relations

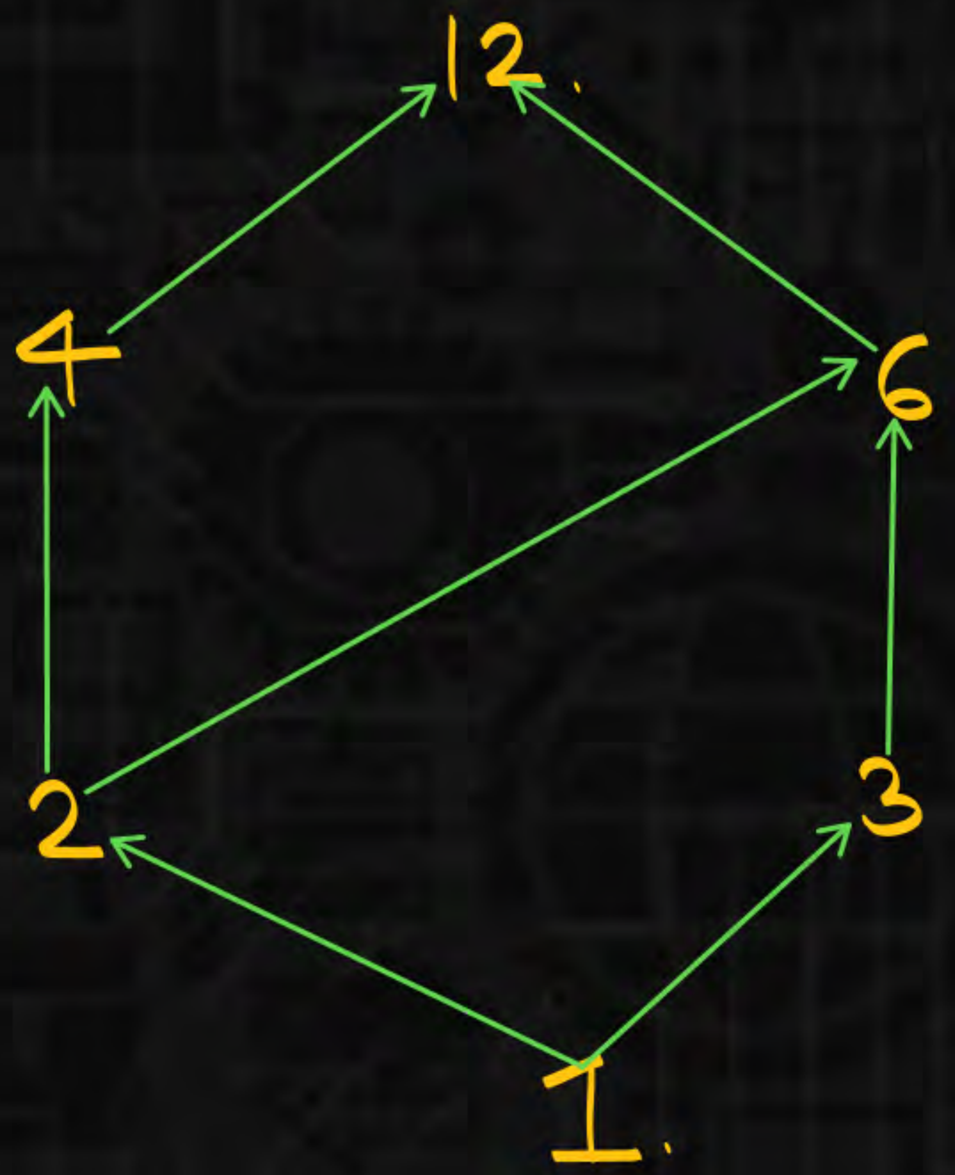
Remove self loop.



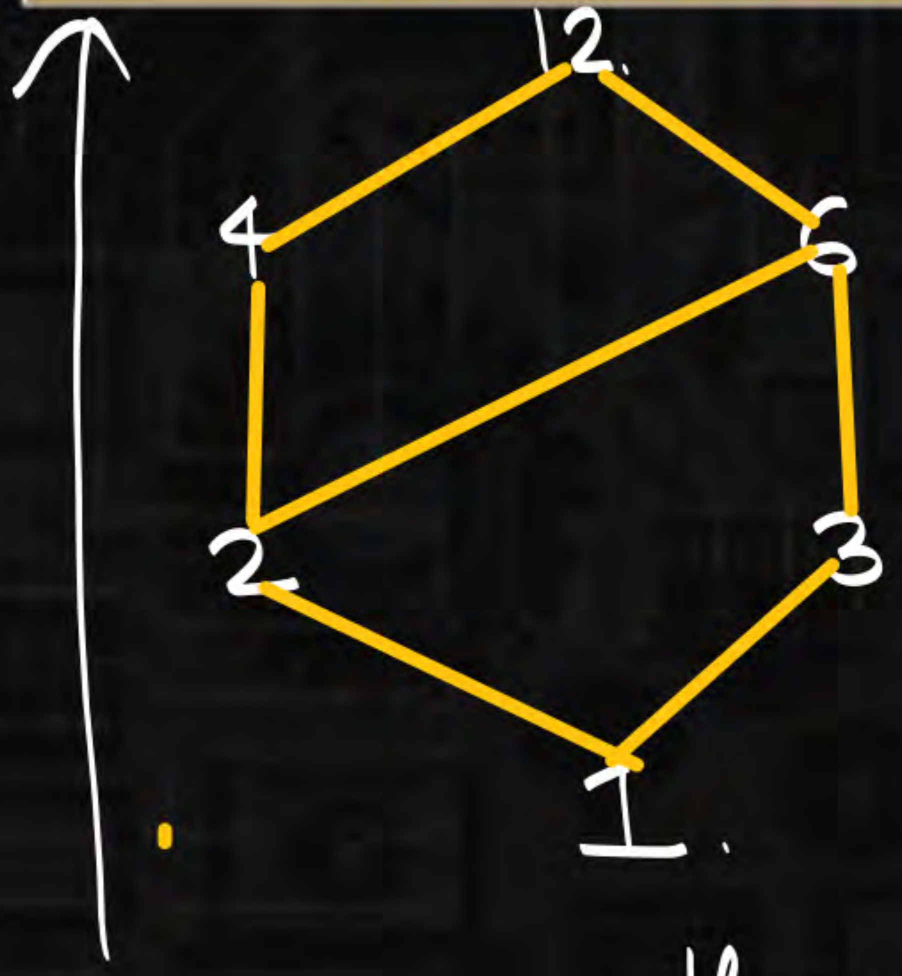
Relations



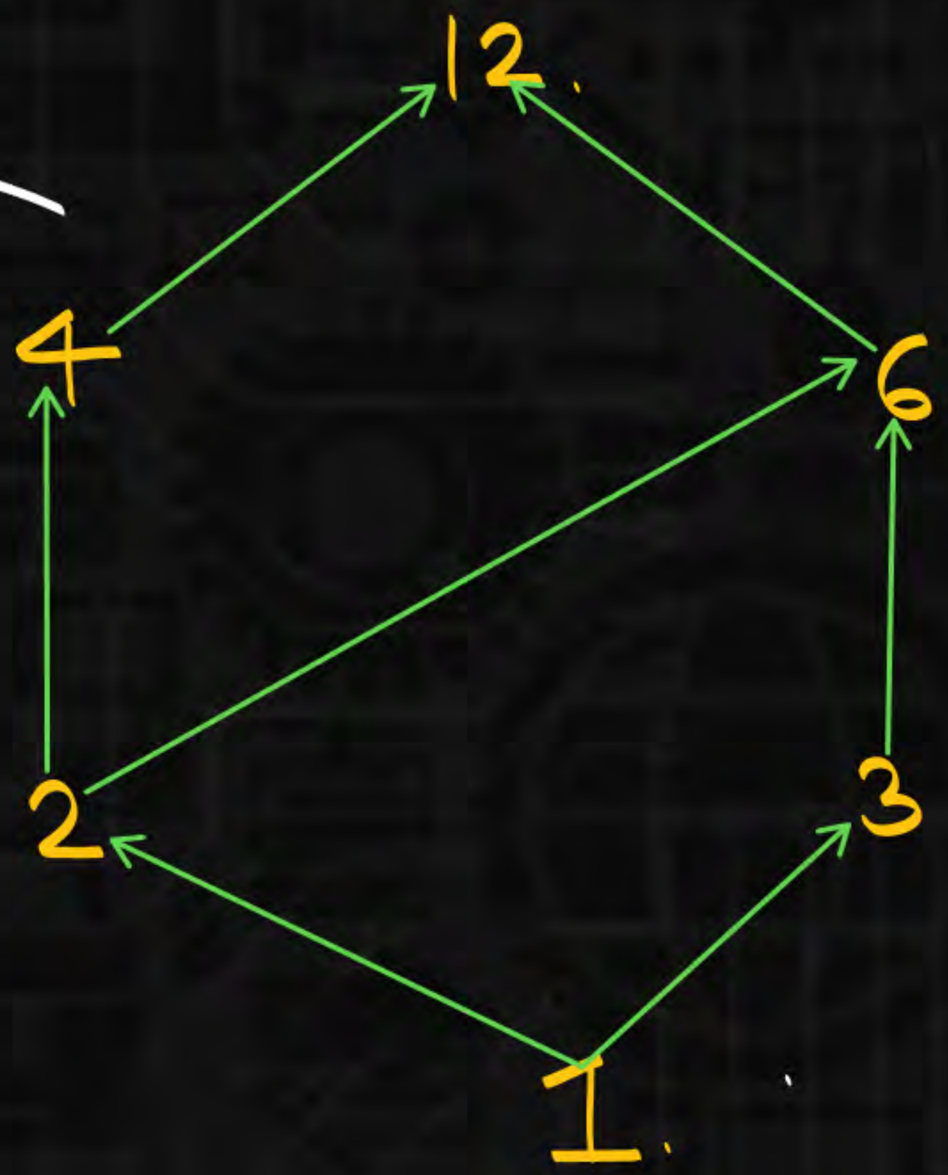
Remove transitive arrows



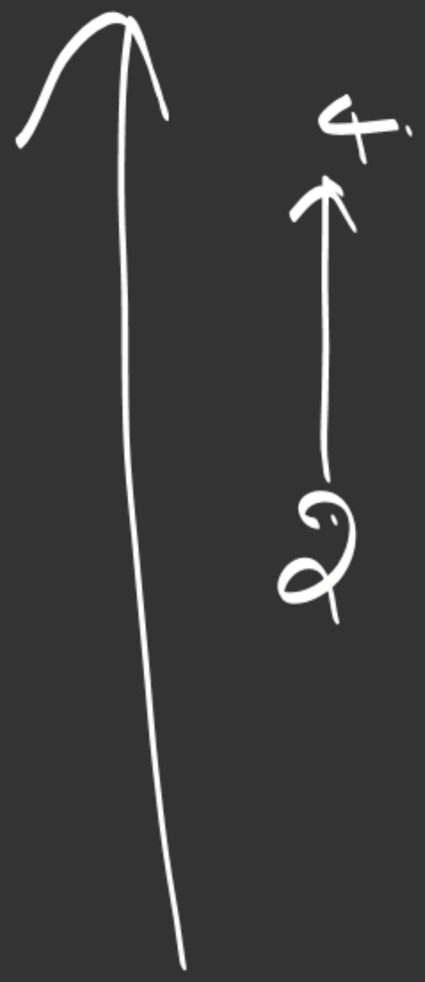
Relations



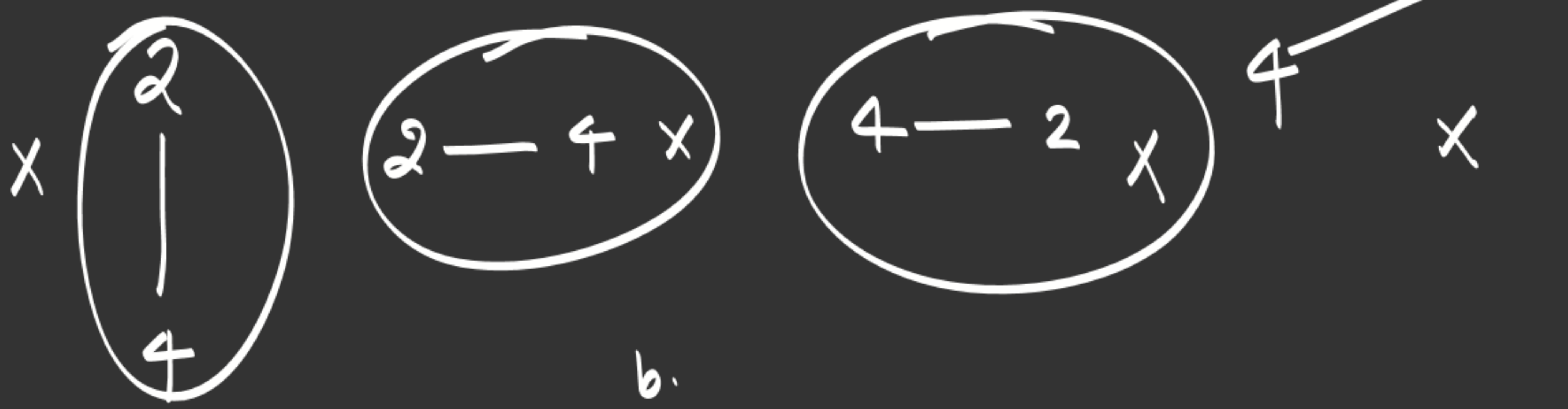
Anti:
Remove
directions



Hasse diagram



Can nodraw



$(a,b) \in R$

$(b,c) \notin R$ b c

Relations

no. of edges in hasse diagram. $(D_{24}, |) \longrightarrow 10$ edges.

$(D_{36}, |) \rightarrow$

$(D_{24}, |)$

$(\{1, 2, 3, 4, 6, 8, 12, 24\}, |)$

Relations

$(D_{24}, |)$

$(\{1, 2, 3, 4, 6, 8, 12, 24\}, |)$

no Relation bet^n 8, 12

so they must be @ same level.

$(D_{36}, |)$

Relations

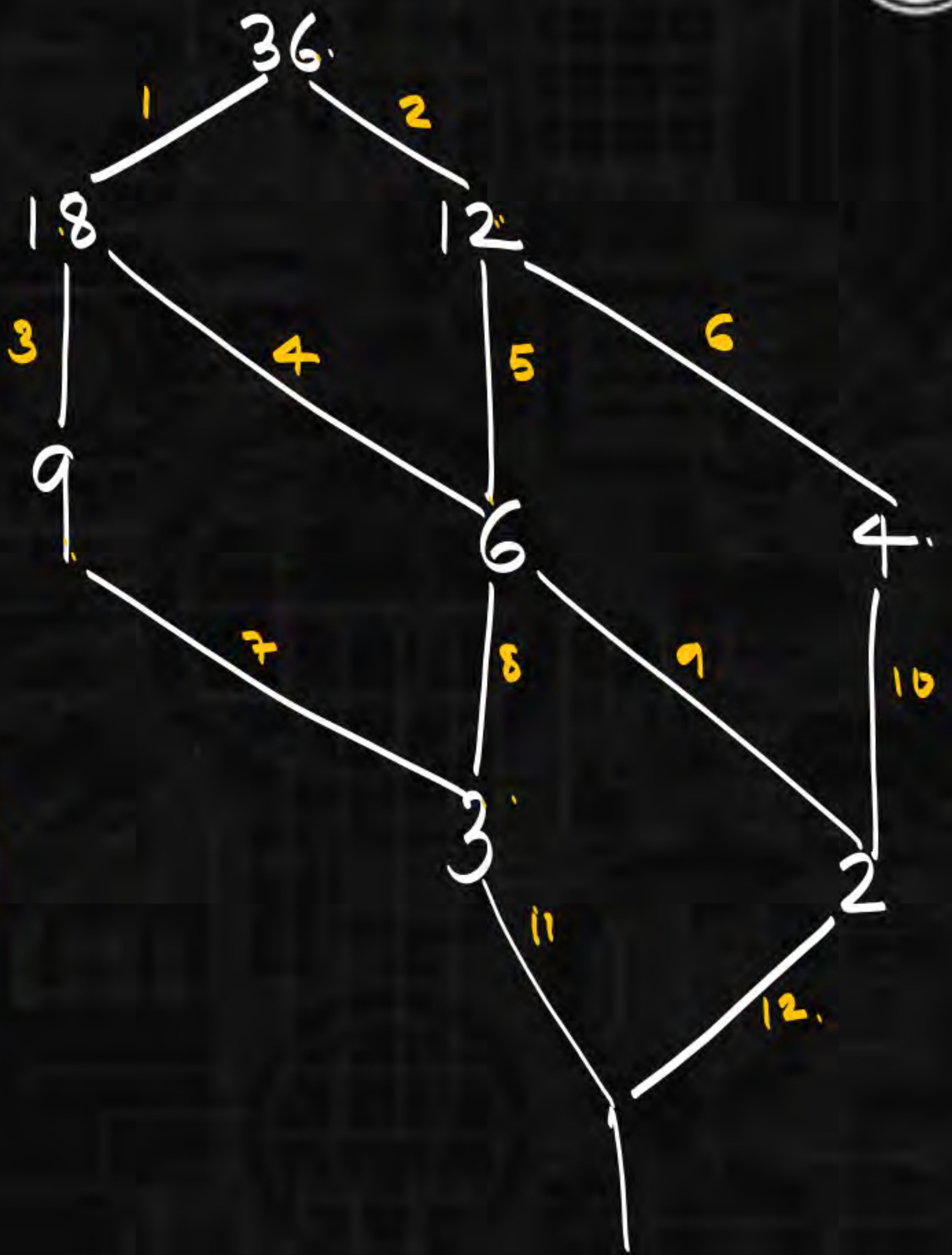
(Set, Relation)

$(D_{36}, |)$

$(\{1, 2, 3, 4, 6, 9, 12, 18, 36\}, |)$

no transitive arrows

12 edges



Relations

H.W :

* 1) Set: $A = \{1, 2, 3\}$

$(P(A), \subseteq)$ Check it is poset?

$(\{\ldots\ldots\}, \subseteq)$

* 2) $A = \{1, 2, 3\}$

$(P(A), \subseteq)$ how many edges will be there in hasse diagram?

** 3) $A = \{1, 2, 3, 4, 5\}$

$(P(A), \subseteq)$

(# of edges in this hasse diagram)

Relations

Set: $A = \{1\ 2\ 3\ 4\ 5\} \times \{1\ 2\ 3\ 4\ 5\}$.

cp. $A^2 \times A^2$.

<u>Relation</u>: $(x_1\ y_1)\,R\,(x_2\ y_2) \mid x_1 + y_1 = x_2 + y_2$.

{Check equivalence Relation.

{Determine its partions ?.

Relations

** $A = \{a, b, c, d, e, f\}$

how many equivalence relation.

if exactly 2 equivalence classes of size 3.


THANK YOU GW
SOLDIERS !